Database Management Assignment File (ENCS205)

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

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B. Tech CSE (Core) – 4th Semester

Section: A

Submitted To:

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1.Consider following databases and draw ER diagram and convert entitiesand relationships to relation table for a given scenario:

Aim: To analyze the requirements for a College Database, design an Entity-Relationship (ER) Diagram representing Students, Sections, Subjects, and Marks, and convert the conceptual model into a Relational Database Schema.

1. Entities and Attributes

1.1 STUDENT

Represents students enrolled in the institution.

Attributes:

USN (Primary Key)

SName

Gender

Address

Phone

Each student is uniquely identified by USN.

1.2 SEMSEC (Semester Section)

Represents a specific semester and section.

Attributes:

SSID (Primary Key)

Sem

Sec

Each semester-section is uniquely identified by SSID.

1.3 SUBJECT

Represents subjects offered in the institution.

Attributes:

SubCode (Primary Key)

Title

Credits

Sem

Each subject is uniquely identified by SubCode.

1.4 IAMARKS (Internal Assessment Marks)

Represents marks obtained by students.

Attributes:

Test1

Test2

Test3

FinalIA

This entity stores internal assessment details of students.

2. Relationships and Cardinality

2.1 CLASS (Between STUDENT and SEMSEC)

A student belongs to one semester-section.

A semester-section contains multiple students.

Cardinality: One-to-Many

2.2 SECURES (Between STUDENT and IAMARKS)

A student secures internal assessment marks.

One student can have multiple IA records (one for each subject).

Cardinality: One-to-Many

2.3 SCORED_IN (Between IAMARKS and SUBJECT)

Marks are scored in a particular subject.

One subject can have marks of many students.

Cardinality: One-to-Many

2.4 CLASS_MARKS (Between SEMSEC and IAMARKS)

Internal assessment records are associated with a semester-section.

One semester-section can have multiple IA records.

Cardinality: One-to-Many

3. Relational Schema

STUDENT(

USN PRIMARY KEY,

SName,

Gender,

Address,

Phone,

SSID FOREIGN KEY

)

SEMSEC(

SSID PRIMARY KEY,

Sem,

Sec

)

SUBJECT(

SubCode PRIMARY KEY,

Title,

Credits,

Sem

)

IAMARKS(

USN FOREIGN KEY,

SubCode FOREIGN KEY,

SSID FOREIGN KEY,

Test1,

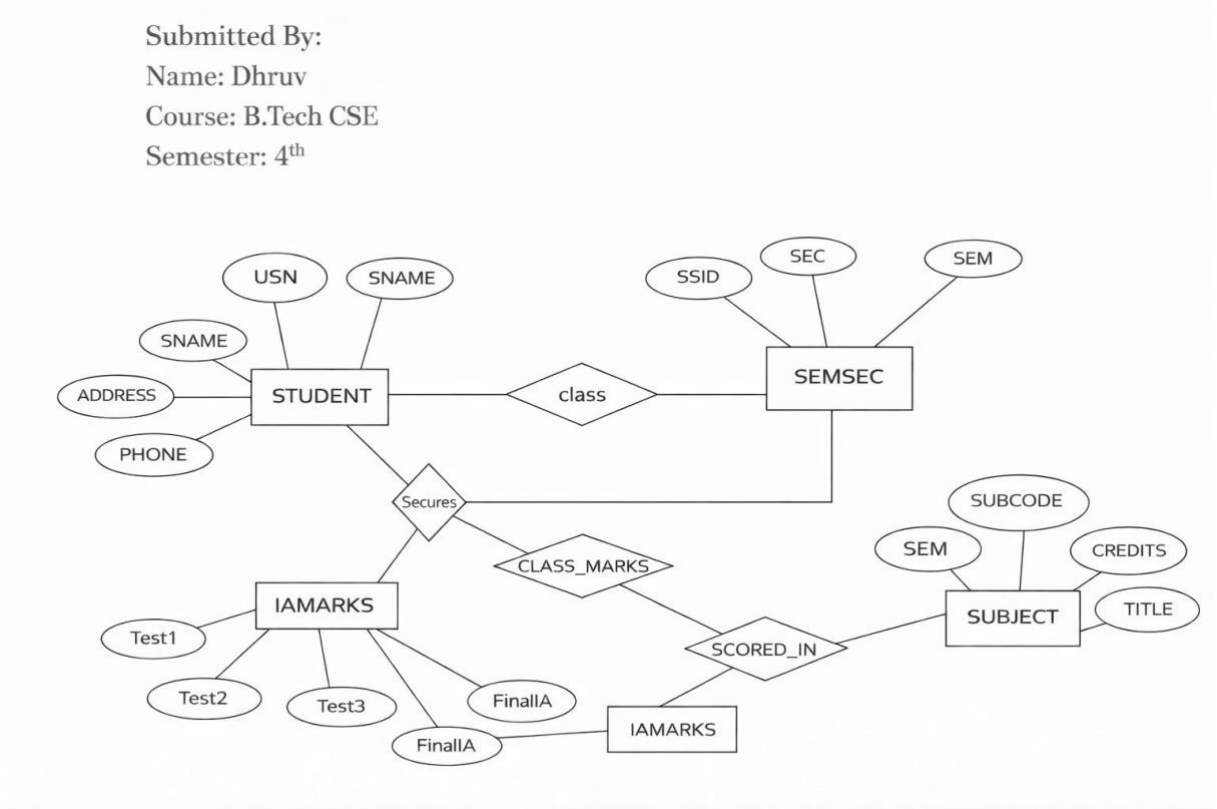
Test2,

Test3,

FinalIA,

PRIMARY KEY (USN, SubCode)

)



2. **Consider following databases and draw ER diagram and convert entities and relationships to relation table for a given scenario:**

AIM: To analyze the requirements for a Company Database, design an Entity-Relationship (ER) Diagram representing Employees, Departments, Projects, and their relationships (Works On, Manages, Supervision), and map the ER diagram into a normalized Relational Database Schema.

Entities and Attributes

1. **EMPLOYEE**

Represents employees working in the organization.

Attributes:

Emp_ID (Primary Key) → 99999

Emp_Name → Dhruv

Dept_ID → D10

Salary → 1000000

Gender → Male

City → New Delhi

2. **DEPARTMENT**

Represents departments in the organization.

Attributes:

Dept_ID (Primary Key) → D10

Location → Delhi

3. **PROJECT**

Represents projects handled by employees.

Attributes:

Project_ID (Primary Key) → F5

Project_Name → EMP Management System

Location → Delhi

Dept_ID → D10

Relationships

1. **Works_For (EMPLOYEE — DEPARTMENT)**

An employee works for one department.

A department can have many employees.

Cardinality: 1 : N

One Department → Many Employees

2. **Located_At (DEPARTMENT — Location)**

A department is located at a particular place.

From diagram: Department located at Delhi.

3**. Working (EMPLOYEE — PROJECT)**

An employee works on a project.

A project can have multiple employees.

Relationship has attributes:

Project_ID → F5

Hours/Code → 155

Emp_ID → 99999

Cardinality: M : N

One Employee → Many Projects

One Project → Many Employees

Relational Schema

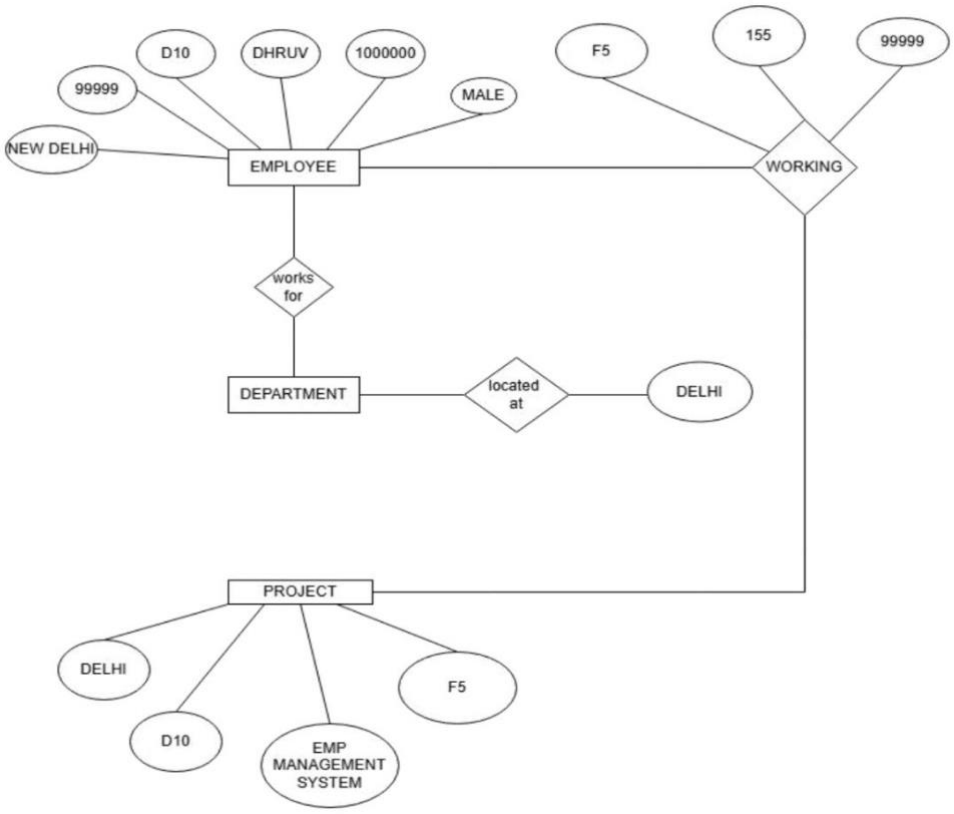
EMPLOYEE(Emp_ID PK, Emp_Name, Dept_ID FK, Salary, Gender, City)

DEPARTMENT(Dept_ID PK, Location)

PROJECT(Project_ID PK, Project_Name, Location, Dept_ID FK)

WORKING(Emp_ID FK, Project_ID FK, Hours,

PRIMARY KEY(Emp_ID, Project_ID))



3.Database Schema for a customer-sale scenario

Customer (Cust id: integer, cust_name: string)

Item (item_id: integer, item_name: string, price: integer)

Sale (bill_no:integer,bill_data:date,cust_id: integer, item_id: integer, qty_sold: integer)

For the above schema, perform the following—

1. Create the tables with the appropriate integrity constraints.
2. Insert around 10 records in each of the tables.
3. List all the bills for the current date with the customer names and item numbers. 4. List the total Bill details with the quantity sold, price of the item and the final amount.
5. List the details of the customer who have bought a product which has a price>200.
6. Give a count of how many products have been bought by each customer
7. Give a list of products bought by a customer having cust_id as 5.
8. List the item details which are sold as of today.

9. Create a view which lists out the bill_no, bill_date, cust_id, item_id, price, qty_sold, amount.
10. Create a view which lists the daily sales date wise for the last one week

```
DROP DATABASE CUSTOMER_SALE;
CREATE DATABASE CUSTOMER_SALE;
USE CUSTOMER_SALE;
CREATE TABLE CUSTOMER(
CUST_ID INT PRIMARY KEY,
CUST_NAME VARCHAR(100)
);
CREATE TABLE ITEM (
ITEM_ID INT PRIMARY KEY,
ITEM_NAME VARCHAR(100),
PRICE INT);
CREATE TABLE SALE (
BILL_NO INT ,
BILL_DATA DATE,
CUST_ID INT,
ITEM_ID INT,
QTY_SOLD INT ,
FOREIGN KEY (CUST_ID) REFERENCES CUSTOMER(CUST_ID),
FOREIGN KEY (ITEM_ID) REFERENCES ITEM(ITEM_ID)
);
INSERT INTO
CUSTOMER(CUST_ID ,CUST_NAME)
VALUES
(1,'Aryan'),
(2,'Harsh'),
(3,'Ankit'),
```

```
(4,'Deepak'),
(5,'Hridyanshu'),
(6,'Ashok'),
(7,'Alok'),
(8,'Aman'),
(9,'Ashish'),
(10,'Mohit');
SELECT * FROM CUSTOMER;

INSERT INTO ITEM VALUES
(101,'Pen',50),
(102,'Notebook',120),
(103,'Bag',800),
(104,'Bottle',250),
(105,'Shoes',1500),
(106,'Watch',2200),
(107,'Mouse',500),
(108,'Keyboard',700),
(109,'Charger',300),
(110,'Headphones',1800);
SELECT * FROM ITEM;


 INSERT INTO SALE VALUES
(1,CURDATE(),1,103,1),
(2,CURDATE(),2,101,3),
(3,CURDATE(),2,105,2),
(4,'2026-02-15',3,104,1),
(5,'2026-02-14',4,106,1),
(6,'2026-02-13',5,102,4),
(7,CURDATE(),6,110,1),
```

```
(8,'2026-02-16',7,108,2),
(9,CURDATE(),5,109,5),
(10,'2026-02-12',8,107,1);
SELECT * FROM SALE; SELECT S.BILL_NO, C.CUST_NAME, S.ITEM_ID
FROM SALE S
JOIN CUSTOMER C ON S.CUST_ID = C.CUST_ID
WHERE S.BILL_DATA = CURDATE();

SELECT
   S.BILL_NO,
   C.CUST_NAME,
   I.ITEM_NAME,
   S.QTY_SOLD,
   I.PRICE,
   (S.QTY_SOLD * I.PRICE) AS FINAL_AMOUNT
FROM SALE S
JOIN CUSTOMER C ON S.CUST_ID = C.CUST_ID
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID;

SELECT DISTINCT C.*
FROM CUSTOMER C
JOIN SALE S ON C.CUST_ID = S.CUST_ID
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID
WHERE I.PRICE > 200;

SELECT C.CUST_NAME, SUM(S.QTY_SOLD) AS TOTAL_PRODUCTS_BOUGHT
FROM CUSTOMER C
JOIN SALE S ON C.CUST_ID = S.CUST_ID
GROUP BY C.CUST_NAME;

SELECT I.ITEM_NAME, S.QTY_SOLD
```

```
FROM SALE S
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID
WHERE S.CUST_ID = 5;
SELECT I.*
FROM SALE S
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID
WHERE S.BILL_DATA = CURDATE();


CREATE VIEW Bill_Details AS
SELECT
   S.BILL_NO,
   S.BILL_DATA,
   S.CUST_ID,
   S.ITEM_ID,
   I.PRICE,
   S.QTY_SOLD,
   (S.QTY_SOLD * I.PRICE) AS AMOUNT
FROM SALE S
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID;

SELECT * FROM BILL_DETAILS;


CREATE VIEW WEEKLY_SALES AS
SELECT
   BILL_DATA,
   SUM(S.QTY_SOLD * I.PRICE) AS TOTAL_DAILY_SALES
FROM SALE S
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID
WHERE BILL_DATA >= CURDATE() - INTERVAL 7 DAY
```

GROUP BY BILL_DATA;

SELECT * FROM WEEKLY_SALES;

OUTPUT

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

CUST_ID	CUST_NAME
1	Aryan
2	Harsh
3	Ankit
4	Deepak
5	Hridyanshu
6	Ashok
7	Alok
8	Aman
9	Ashish
10	Mohit
NULL	NULL

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

ITEM_ID	ITEM_NAME	PRICE
101	Pen	50
102	Notebook	120
103	Bag	800
104	Bottle	250
105	Shoes	1500
106	Watch	2200
107	Mouse	500
108	Keyboard	700
109	Charger	300
110	Headphones	1800
NULL	NULL	NULL

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

BILL_NO	BILL_DATA	CUST_ID	ITEM_ID	QTY_SOLD
1	2026-02-28	1	103	1
2	2026-02-28	2	101	3
3	2026-02-28	2	105	2
4	2026-02-15	3	104	1
5	2026-02-14	4	106	1
6	2026-02-13	5	102	4
7	2026-02-28	6	110	1
8	2026-02-16	7	108	2
9	2026-02-28	5	109	5
10	2026-02-12	8	107	1

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

BILL_NO	CUST_NAME	ITEM_ID
1	Aryan	103
2	Harsh	101
3	Harsh	105
7	Ashok	110
9	Hridyanshu	109

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

BILL_NO	CUST_NAME	ITEM_NAME	QTY_SOLD	PRICE	FINAL_AMOUNT
1	Aryan	Bag	1	800	800
2	Harsh	Pen	3	50	150
3	Harsh	Shoes	2	1500	3000
4	Ankit	Bottle	1	250	250
5	Deepak	Watch	1	2200	2200
6	Hridyanshu	Notebook	4	120	480
7	Ashok	Headphones	1	1800	1800
8	Alok	Keyboard	2	700	1400
9	Hridyanshu	Charger	5	300	1500
10	Aman	Mouse	1	500	500

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CUST_ID	CUST_NAME
1	Aryan
3	Ankit
2	Harsh
4	Deepak
8	Aman
7	Alok
5	Hridyanshu
6	Ashok

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CUST_NAME	TOTAL_PRODUCTS_BOUGHT
Aryan	1
Harsh	5
Ankit	1
Deepak	1
Hridyanshu	9
Ashok	1
Alok	2
Aman	1

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ITEM_NAME	QTY_SOLD
Notebook	4
Charger	5

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ITEM_ID	ITEM_NAME	PRICE
103	Bag	800
101	Pen	50
105	Shoes	1500
110	Headphones	1800
109	Charger	300

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

BILL_NO	BILL_DATA	CUST_ID	ITEM_ID	PRICE	QTY_SOLD	AMOUNT
1	2026-02-28	1	103	800	1	800
2	2026-02-28	2	101	50	3	150
3	2026-02-28	2	105	1500	2	3000
4	2026-02-15	3	104	250	1	250
5	2026-02-14	4	106	2200	1	2200
6	2026-02-13	5	102	120	4	480
7	2026-02-28	6	110	1800	1	1800
8	2026-02-16	7	108	700	2	1400
9	2026-02-28	5	109	300	5	1500
10	2026-02-12	8	107	500	1	500

800

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

BILL_DATA	TOTAL_DAILY_SALES
2026-02-28	7250